江苏省大学生万人计划
学术冬令营

南京大学

基于机器视觉的农业病虫害识别研究

小组成员
杨立颖 王清玉
马　鑫　邱铮佳

汇报人
张 伟

TERM，TERM POWER



江苏省大学生万人计划
学术冬令营

目录
CONTENT

AI+农业，下一场“工业革命”即将爆发

选题背景 PART ONE

研究现状 PART TWO

实现方案 PART THREE

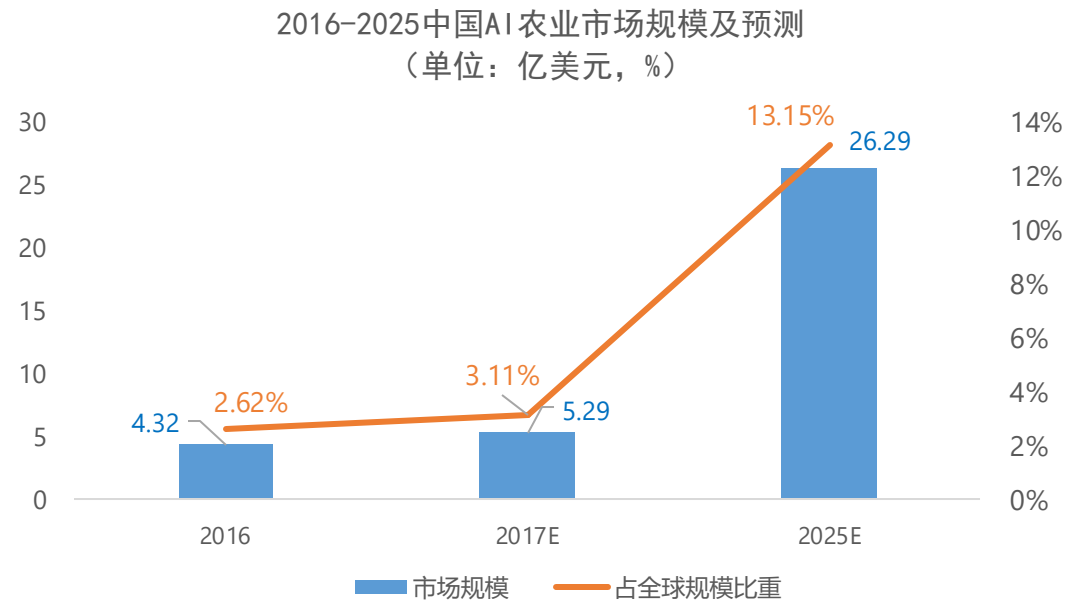
创新应用 PART FOUR

参考文献 PART FIVE

选题背景
PART ONE

农业问题亟待解决

植物病虫害检测预警是防灾减灾工作的关键之一，联合国粮农组织的研究表明，进农作物病虫害危害自然损失率就超过37%，中国植物病虫害种类多，发生面积广，危害严重，暴发频率逐年提高，损失逐年加重，若不采取防控措施，因病虫危害中国每年将损失粮食1500亿Kg、果品与蔬菜1000亿Kg、油料68亿Kg、棉花1.9亿Kg，潜在经济损失在5000亿元以上。（来自《农药市场信息》）



数据来源：前瞻产业研究院整理

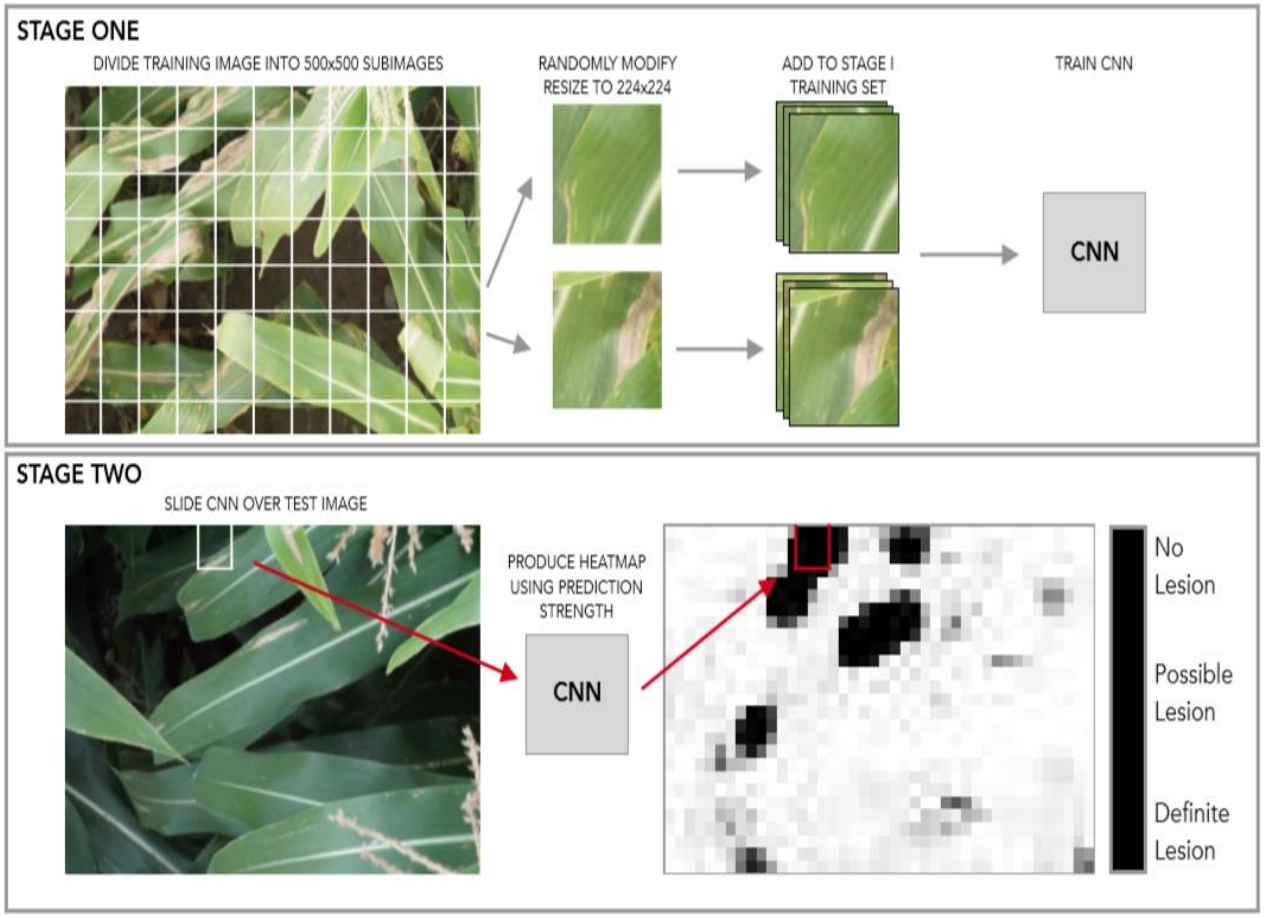
- ◆虫害严重影响林业生态环境和农作物产量
- ◆目前识别病虫害的自动化程度不高，大多依靠人眼手动识别
- ◆近年来，人工智能和深度学习技术得到快速发展与应用
- ◆无人机航拍技术在防治病虫害领域发挥着越来越重要的作用





研究现状
PART TWO

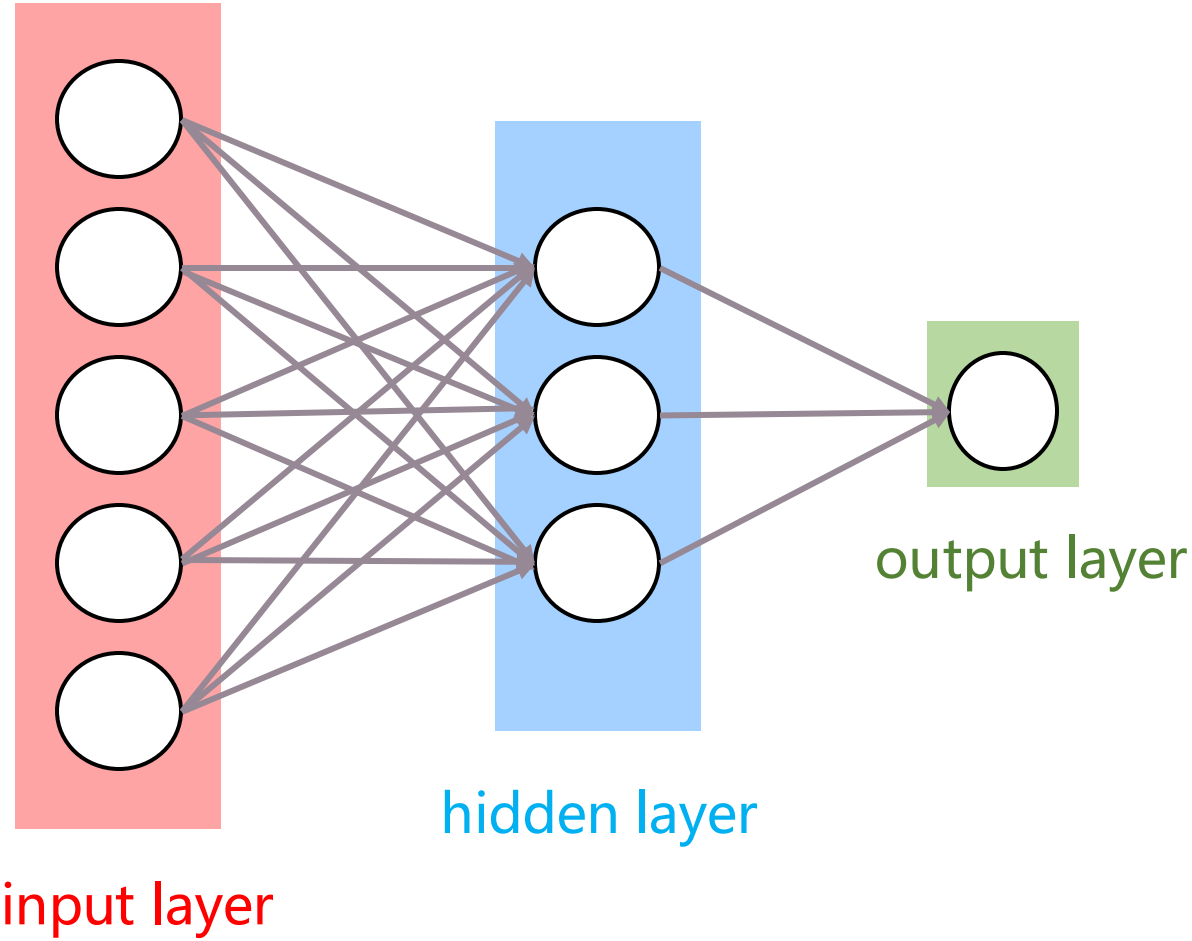
Harvey Wu等人采用小型无人机系统获取田间高分辨率图像，在较低分辨率的子图像上训练卷积神经网络（CNN）模型。在与训练过程分离的一组子图像上识别的准确率高达**95%**。





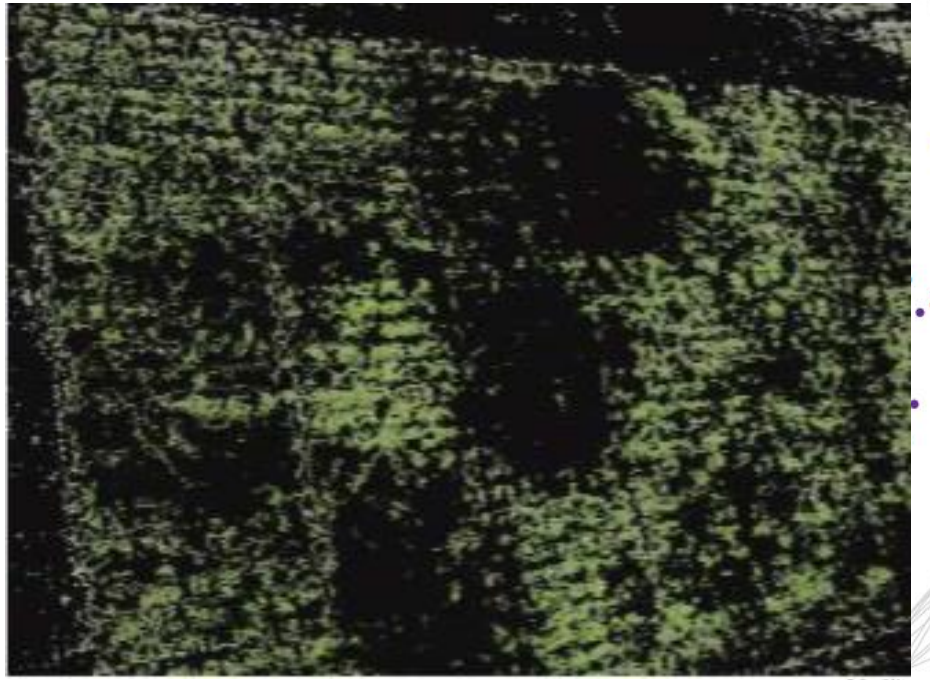


Alexander A. DOUDKIN等人采用航拍的作物RGB彩色图像，通过**反向传播训练的三层集群化神经网络算法**，识别作物病虫害区域并进行分割每种疾病训练一个单独的神经网络。



识别效果图

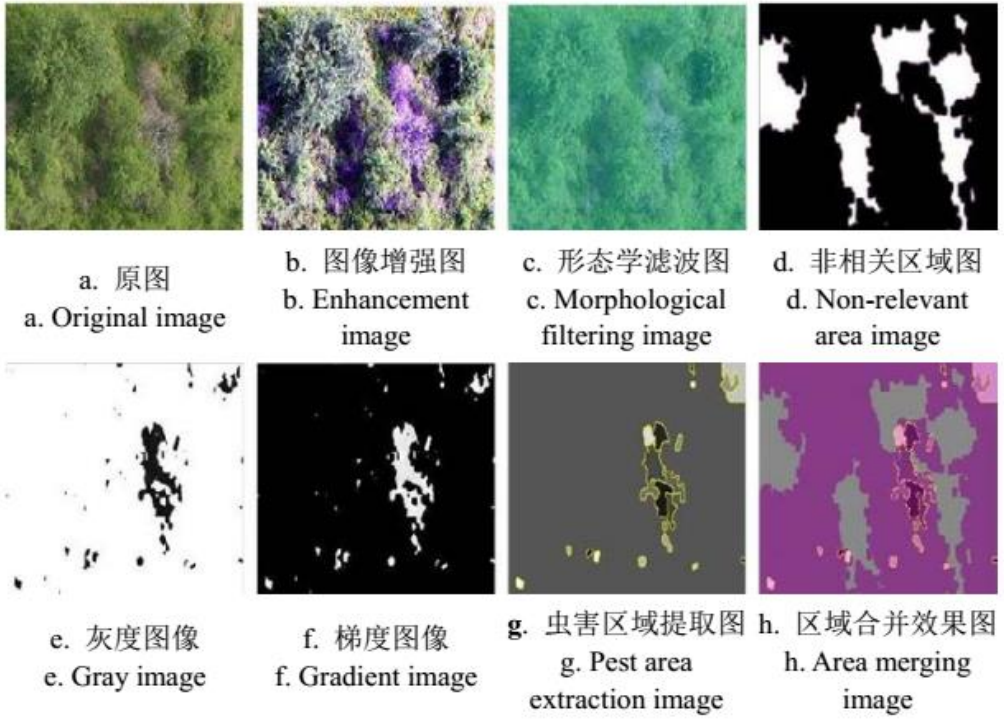




张军国等人提出了**复合梯度分水岭算法**，是一种分水岭变换对梯度图像进行分割提取虫害区域并进行区域合并的方法。

缺点 **相对误差率**以及测量精度方面还是偏低；

对非规则区的识别能力也不是很强。



a. 原图 a. Original image

b. 图像增强图 b. Enhancement image

c. 形态学滤波图 c. Morphological filtering image

d. 非相关区域图 d. Non-relevant area image

e. 灰度图像 e. Gray image

f. 梯度图像 f. Gradient image

g. 虫害区域提取图 g. Pest area extraction image

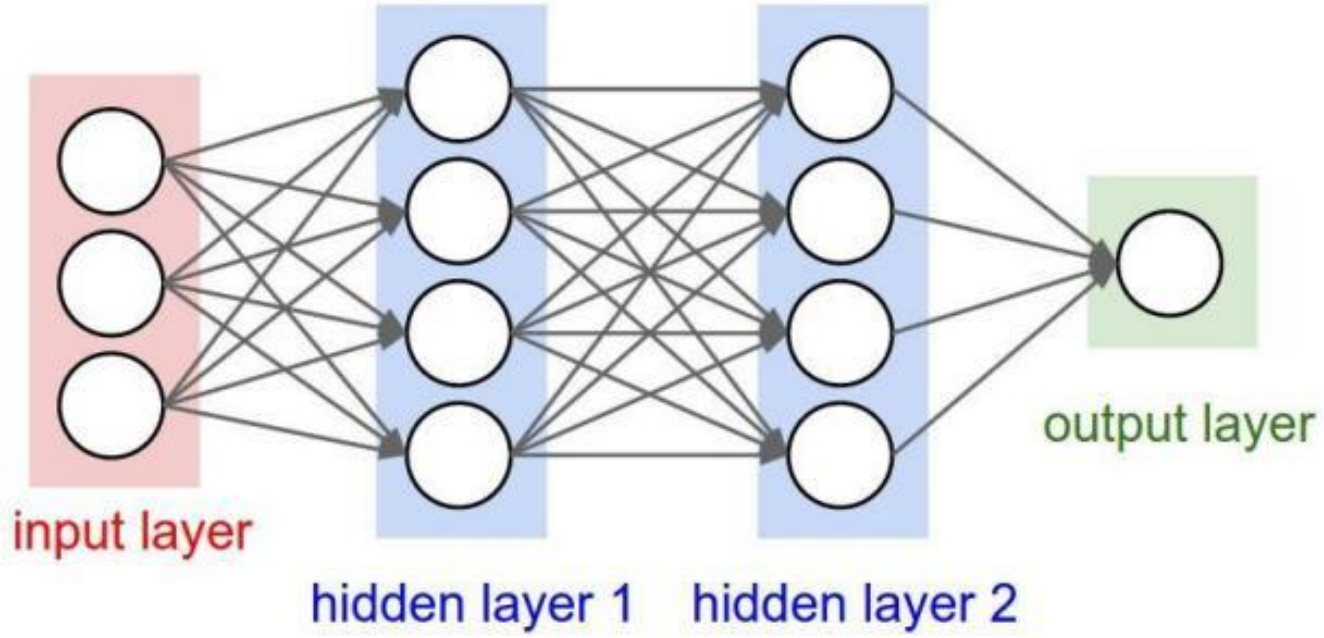
h. 区域合并效果图 h. Area merging image

序号 No.	传统分水岭算法 Traditional watershed algorithm		*K*-means 聚类算法 *K*-means clustering algorithm		本文算法 Our algorithm	
	相对误差率	相对极限测量精度	相对误差率	相对极限测量精度	相对误差率	相对极限测量精度
1	8.43	7.32	3.61	3.71	1.49	1.12
2	7.62	9.14	5.28	4.74	1.55	0.92
3	9.12	8.92	4.32	4.21	1.91	0.87
4	7.38	9.11	3.57	3.27	1.62	1.62
5	8.18	7.48	5.67	3.42	1.63	1.68
6	7.38	8.93	5.83	4.30	1.54	1.10
7	8.12	7.45	4.23	2.82	1.37	1.47
8	9.32	9.64	5.87	3.25	1.94	1.72
均值 Mean value	8.19	8.50	4.80	3.72	1.63	1.31

实施方案

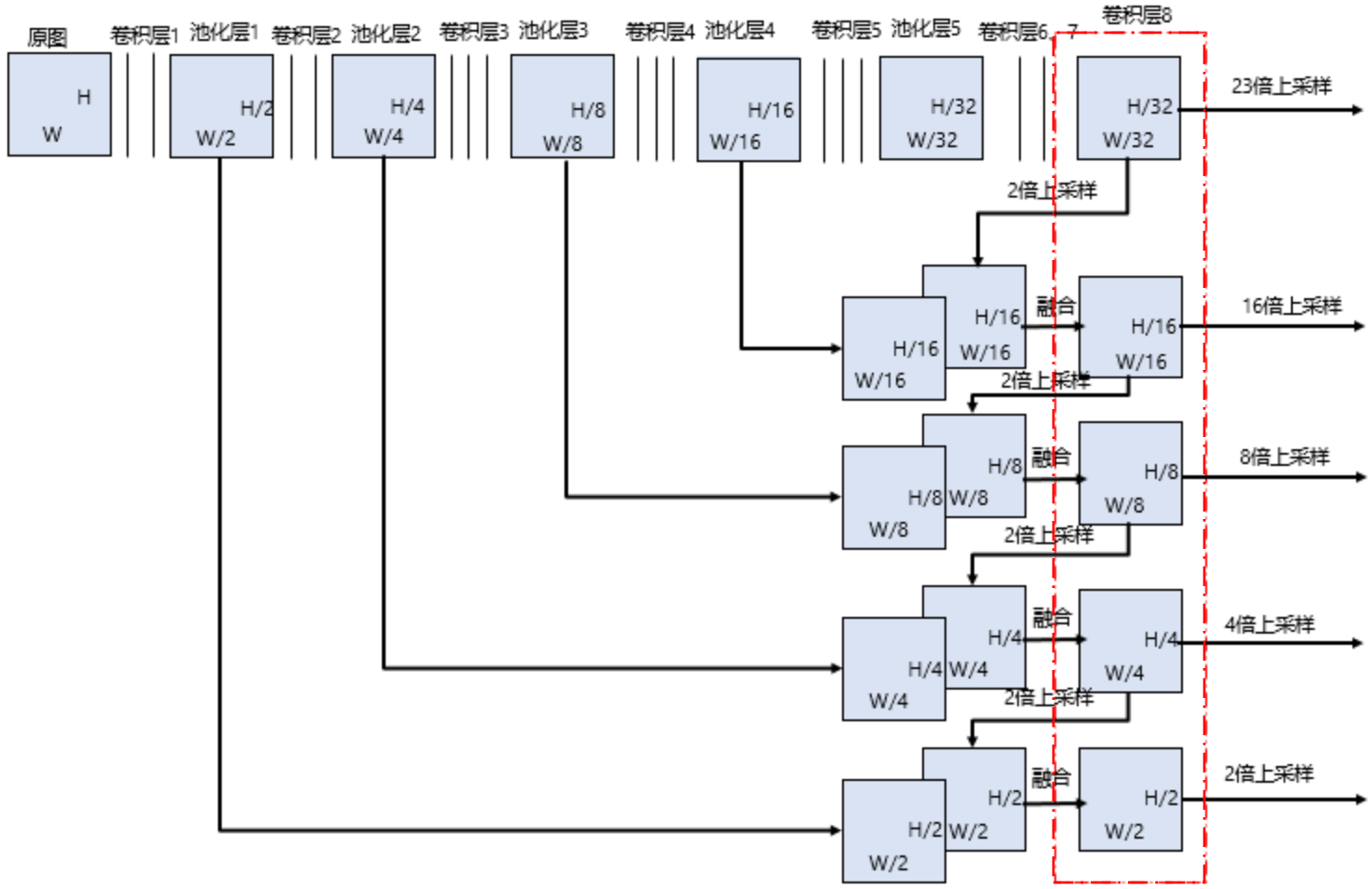
PART THREE

基本原理



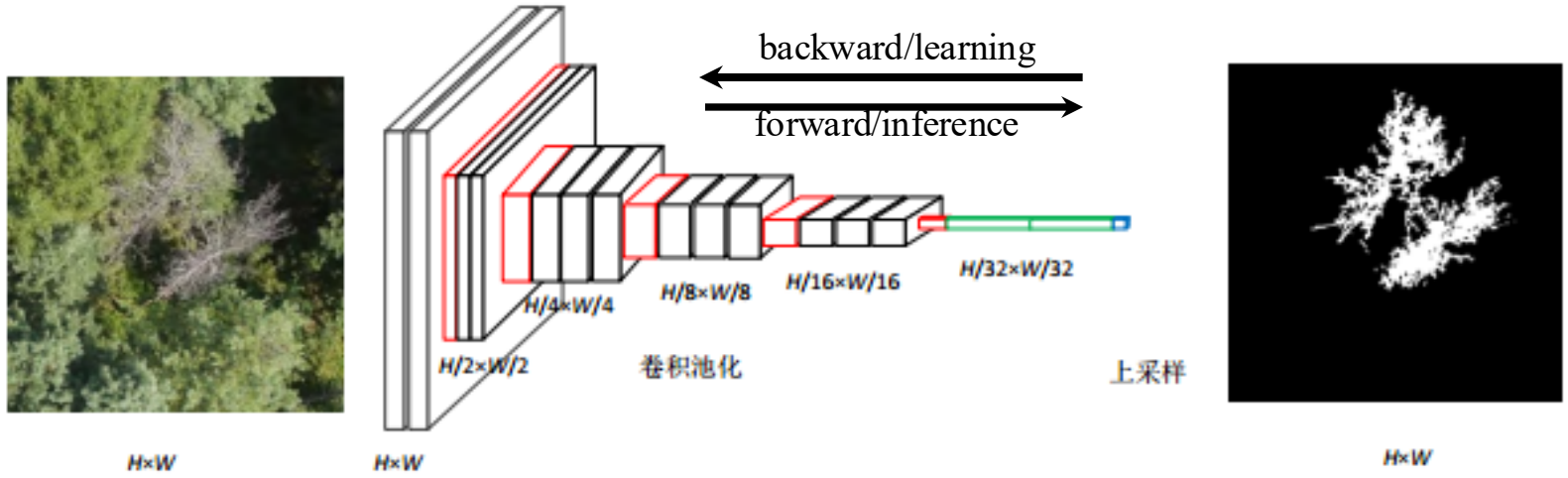
传统卷积神经网络结构图

基本原理



全卷积网络跳跃结构示意图

预实施方案

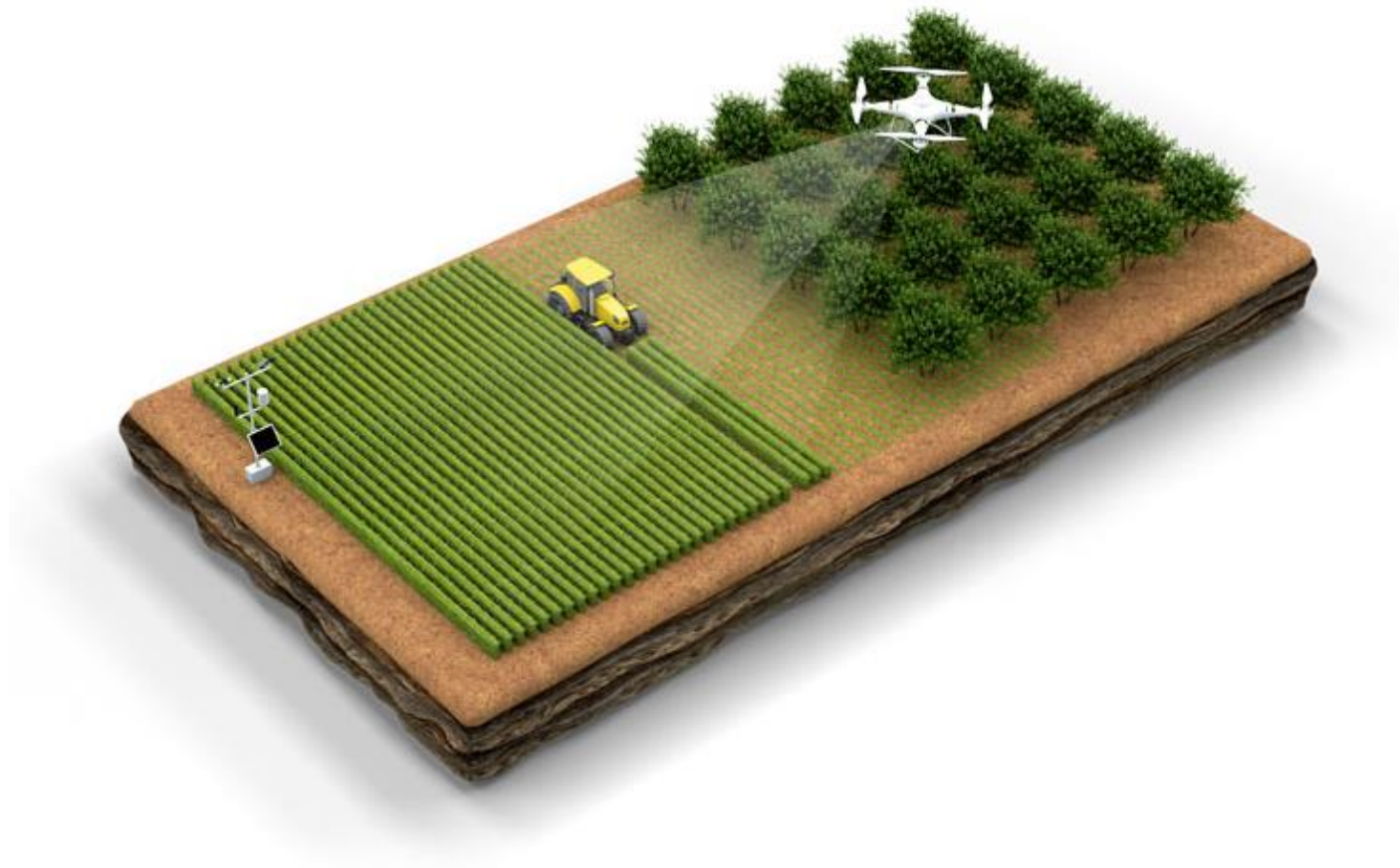


图中H为图像高度，W为图像宽度

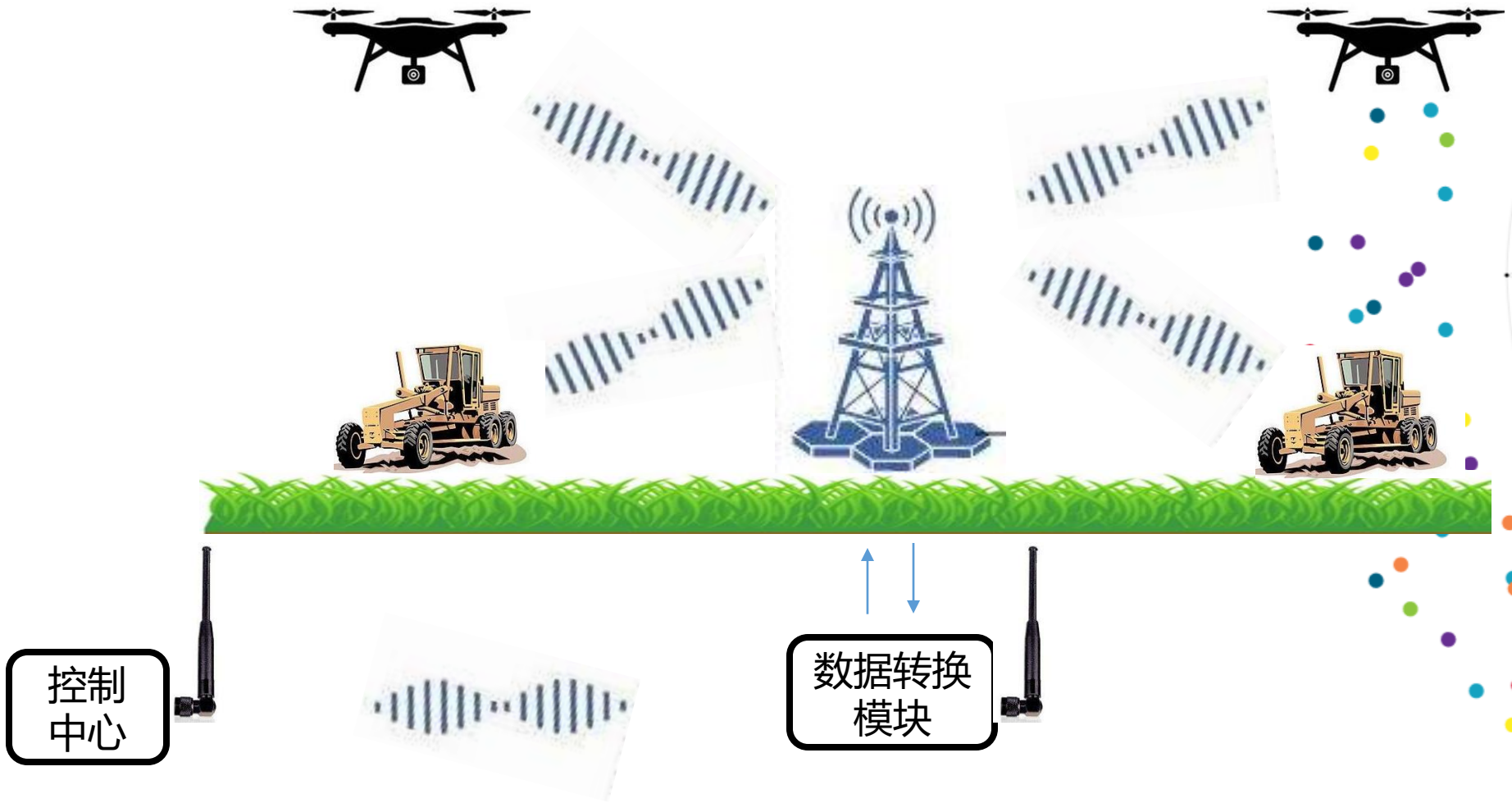
测试结果

算法	识别精度评估指标			
	像素准确率	类别平均准确率	平均交并比	频率加权平均交并比
K-means 51.85	52.93	59.07	29.30	49.21
PCNN	77.13	47.88	43.82	75.21
CGWA	91.82	84.38	60.63	88.71
FCN32s	96.76	79.22	71.86	94.20
FCN16s	97.24	79.88	74.11	94.96
FCN 8s	97.46	87.55	77.51	95.49
FCN 4s	97.52	87.85	78.12	95.58
FCN 2s	97.86	87.12	79.49	96.11

创新应用
PART FOUR



精准农业





参考文献
PART FIVE

PART FIVE 参考文献

[1] Wu H, Wiesner-Hanks T, Stewart E L, et al. Autonomous End-to-end Detection of Northern Leaf Blight directly from Aerial Imagery[J].

[2] Doudkin A A, Inyutin A V, Petrovsky A I, et al. Three-level neural network for data clusterization on images of infected crop field[J]. Journal of Research and Applications in Agricultural Engineering, 2007, 52(1): 5-7.

[3]孙钰，韩京冶，陈志泊，等. 基于深度学习的大棚及地膜农田无人机航拍监测方法[J/OL]. 农业机械学报, 2018,49(2): 133-140.

[4]刘文定，田洪宝，谢将剑，赵恩庭，张军国．基于全卷积网络的林业航拍图像虫害区域识别方法[J/OL]．农业机械学报. http://kns.cnki.net/kcms/detail/11.1964.s.20181226.1637.018.html


农业机械学

基于全卷积网络的林业航拍图

刘文定[1,2] 田洪宝[1,2] 谢将

(1. 北京林业大学工学院, 北京 100083; 2. 林业装备与

摘要：针对航拍林区虫害图像的虫害区域不规则和传统识别方法

networks,FCN)的虫害区域识别方法。采用八旋翼无人机航拍虫

型训练，将VGG16模型的全连接层替换为卷积层并通过上采样

采用跳跃结构融合多种特征信息，有效提升识别精度，并通过

在5种全卷积网络中区域识别精度最高，其识别结果的像素

与k-means、脉冲耦合神经网络、复合梯度分水岭算法相比

出50.19、35.67和18.86个百分点，单幅分割时间分别降

准确识别，为林区虫害监测和防治提供基础。

关键词：林业虫害监测；航拍；图像识别；全卷积网

中图分类号：TP79；S763.3 （遥感技术的应用：虫

Identification Methods for Based on

LIU Wending[1,2] TIAN Hongb

(1. *School of Tec*

2. *Key Laboratory of State Forestry and Gr*

Abstract: Aiming at the problem of ir forestry pest images in the following pest image segmentation based on f of pest area. First of all, the insect photograph technique over the pe we replaced the full connection l implementing upsampling; ther speed of the model; finally, v recognition accuracy, and fiv had the highest recognition accuracy of the segmentati single image is 4.31s. C algorithm, its pixel accur above 50.19%, 35.67% respectively. This met basis for pest detectio...

收稿日期：2018-09-14 修回日期：2018-11-12

基金项目：中央高校基本科研业务费专项资金项目(2016ZCQ09)

作者简介：刘文定（1968—），女，教授，主要从事智能控制研究。E-mail: liuwending@...

通信作者：张军国（1978—），男，教授，博士生导师，主要从事图像处理与人工智能研究。E-mail: ...

PLOS ONE

RESEARCH ARTICLE

Detection of Laurel Wilt Disease in Avocado Using Low Altitude Aerial Imaging

Ana L. de Castro[1], Reza Ehsani[1]*, Randy C. Ploetz[2], Jonathan H. Crane[2], Sherrie Buchanon[1]

1 Citrus Research and Education Center, University of Florida/IFAS, 700 Experiment Station Road, Lake Alfred, FL 33850, United States of America, 2 Tropical Research and Education Center, University of Florida/IFAS, 18905 SW 280th St, Homestead, 33031 Florida, United States of America

* ehsani@ufl.edu

Abstract

Laurel wilt is a lethal disease of plants in the Lauraceae plant family, including avocado (Persea americana). This devastating disease has spread rapidly along the southeastern seaboard of the United States and has begun to affect commercial avocado production in Florida. The main objective of this study was to evaluate the potential to discriminate laurel wilt-affected avocado trees using aerial images taken with a modified camera during helicopter surveys at low-altitude in the commercial avocado production area. The ability to distinguish laurel wilt-affected trees from other factors that produce similar external symptoms was also studied. $R_{mod}GB$ digital values of healthy trees and laurel wilt-affected trees, as well as fruit stress and vines covering trees were used to calculate several vegetation indices (VIs), band ratios, and VI combinations. These indices were subjected to analysis of variance (ANOVA) and an *M*-statistic was performed in order to quantify the separability of those classes. Significant differences in spectral values among laurel wilt affected and healthy trees were observed in all vegetation indices calculated, although the best results were achieved with Excess Red (ExR), (Red–Green) and Combination 1 (COMB1) in all locations. B/G showed a very good potential for separate the other factors with symptoms similar to laurel wilt-affected trees, such as fruit stress and vines covering trees, from laurel wilt-affected trees. These consistent results prove the usefulness of using a modified camera ($R_{mod}GB$) to discriminate laurel wilt-affected avocado trees from healthy trees, as well as from other factors that cause the same symptoms and suggest performing the classification in further research. According to our results, ExR and B/G should be utilized to develop an algorithm or decision rules to classify aerial images, since they showed the highest capacity to discriminate laurel wilt-affected trees. This methodology may allow the rapid detection of laurel wilt-affected trees using low altitude aerial images and be a valuable tool in mitigating this important threat to Florida avocado production.

OPEN ACCESS

Citation: de Castro AI, Ehsani R, Ploetz RC, Crane JH, Buchanon S (2015) Detection of Laurel Wilt Disease in Avocado Using Low Altitude Aerial Imaging. PLoS ONE 10(4): e0124642. doi:10.1371/journal.pone.0124642

Academic Editor: Jeun Kyu Hong, Gyeongsang National University of Science and Technology, KOREA, REPUBLIC OF

Received: September 17, 2014

Accepted: March 17, 2015

Published: April 30, 2015

Copyright: © 2015 de Castro et al. This is an open access article distributed under the terms of the Creative Commons Attribution License, which permits unrestricted use, distribution, and reproduction in any medium, provided the original author and source are credited.

Data Availability Statement: All relevant data are within the paper.

Funding: Florida Department of Agriculture and Consumer Services, USDA Specialty Block Grant No. 019730, funded this investigation.

Competing Interests: The authors have declared that no competing interests exist.



WE ARE A TERM

学术无止境，百花齐放，奇光异彩！

江苏省大学生万人计划
学术冬令营

南京大学

THANK YOU FOR LISTENING

小组成员
杨立颖 王清玉
马　鑫 邱铮佳

汇报人
张 伟

TERM，TERM POWER